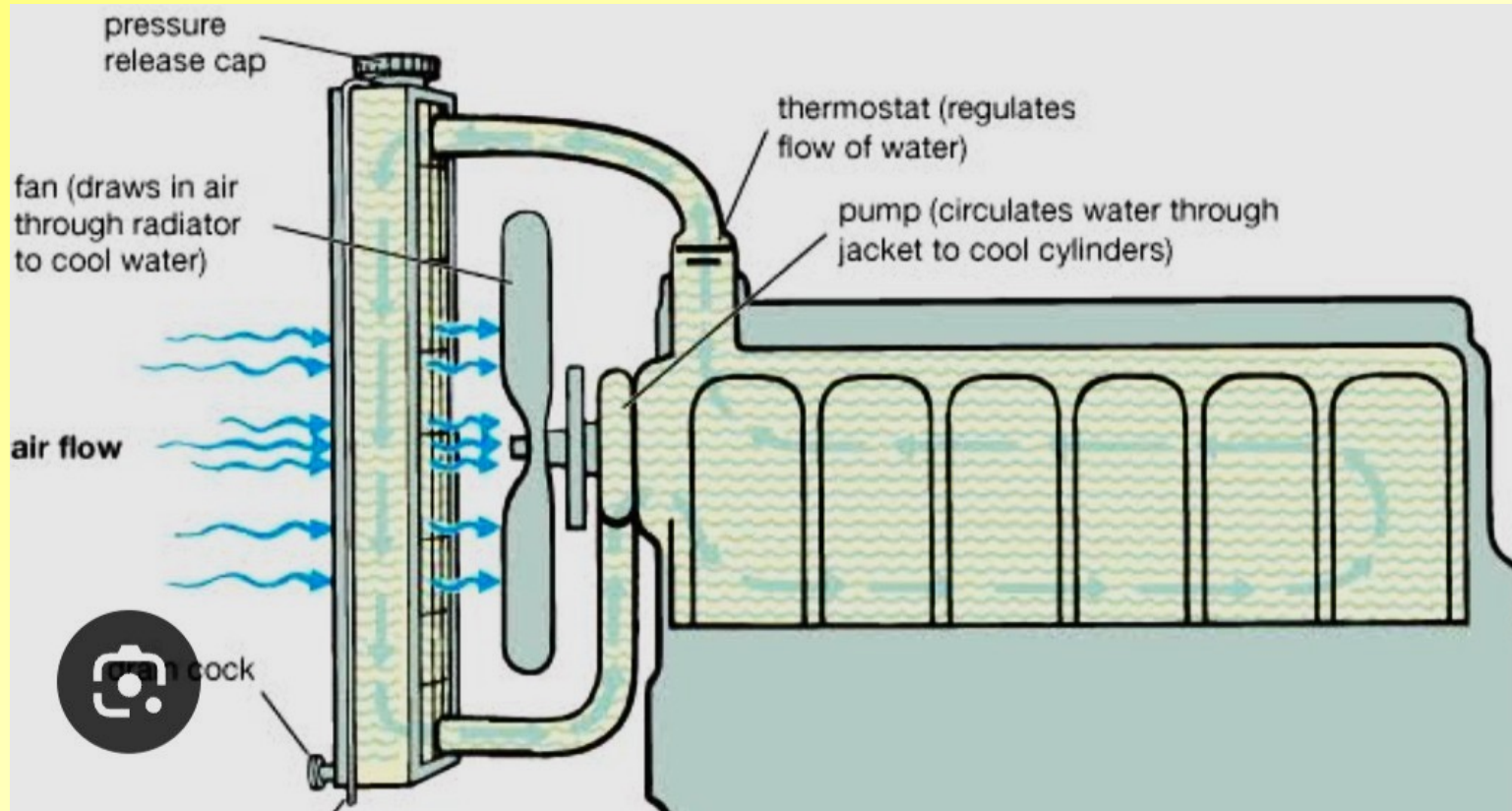




COOLING OF I. C. ENGINES



Dr P M Meena

(PhD: IIT-Bombay)

Professor

Department of Mechanical Engineering

Faculty of Engineering & Architecture

MBM University, Jodhpur - 342001 (Rajasthan)



SYLLABUS



S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.

Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.

Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.



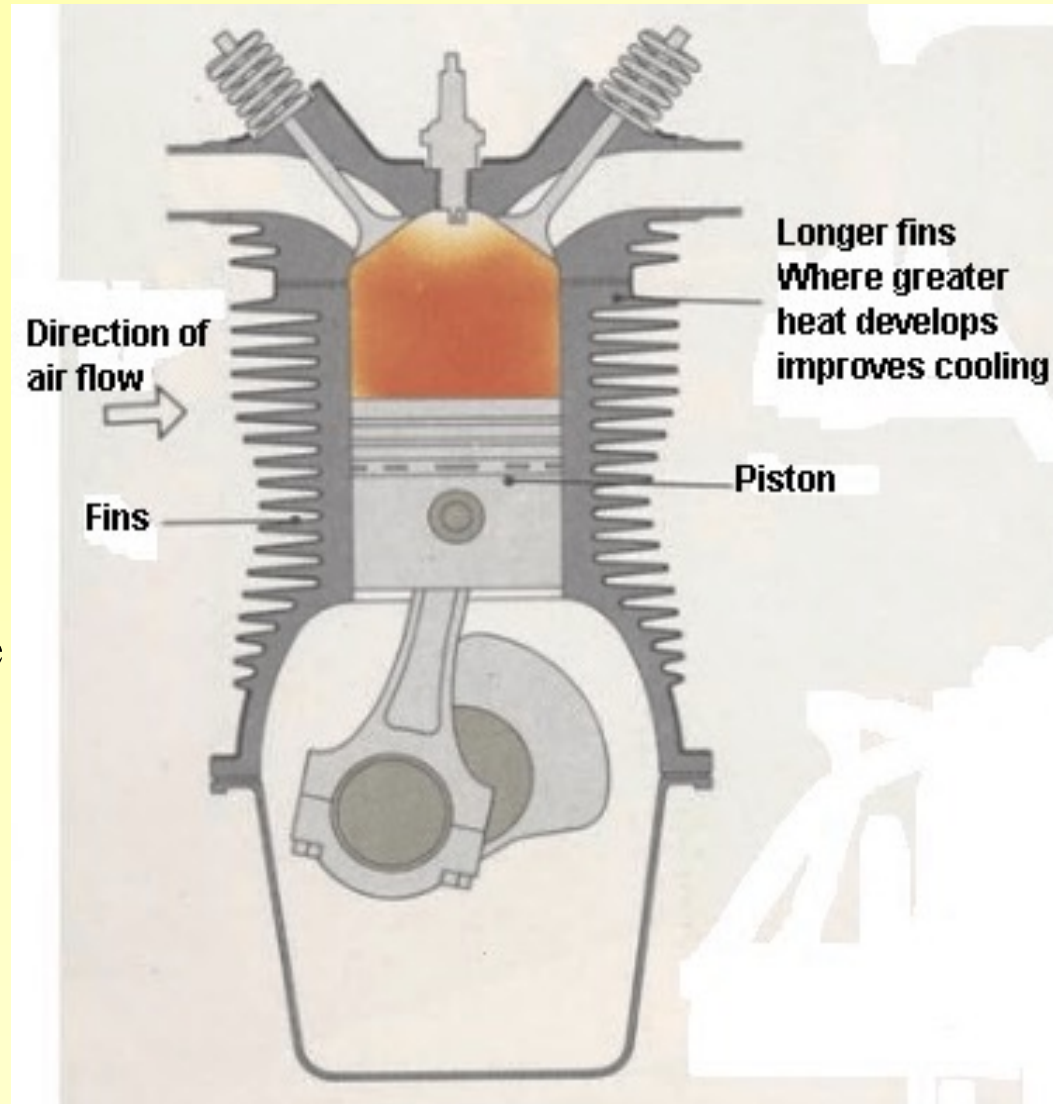
COOLING OF I.C. ENGINES

- *Cooling systems in Internal Combustion Engines play a fundamental role in maintaining the **Efficient Operating temp** by Air or Liquid cooling to remove waste heat and keep engine cool.*
- *In I C Engines,*
 - ✓ Combustion of air - fuel mixture take place inside the engine cylinder
 - ✓ Generated hot gases of about 500-700°C temperature.
- *This is a very high temperature and may result in*
 - ✓ Burning of lubricating **oil-film** between the moving parts,
 - ✓ **Seizure**, sticking or **welding** of the mating parts and
 - ✓ **Overheating** of engines prone to reduce **performance** and **lifespan**.
- *So, temp must be in the range of about **150-200°C** for **efficient** working.*
 - ✓ Too much cooling also **reduces** the work-output and thermal efficiency of engine.
- *The cooling system, consisting of*
 - ✓ Various **components** and **fluids**, are
 - ✓ Responsible for dissipating combustion heat ensuring a **safe** operation temperature range.
- *There are mainly two engine cooling systems*
 - ✓ (a) Air cooled systems, and
 - ✓ (b) Water cooled systems.
- *Most ICs are fluid cooled using either **air** or **liquid** coolant run through 'radiator' cooled by air.*



AIR COOLING SYSTEM

- **Working:** it works by expanding the surface area or by increasing air flow rate or both over the object.
 - ✓ **Fins** cast on cylinder walls, cylinder head etc.
 - ✓ Heat conducted by fins and **dissipated** to air.
- **Amount of heat dissipation depend on the**
 - ✓ **Amount** of air flow-rate over the fin-surfaces
 - ✓ **Thermal conductivity** of the fin material
- **Advantages**
 - ✓ Simple, light due to absence of radiator/pump
 - ✓ Leakage free being air as cooling media,
 - ✓ No coolant freezing problem/antifreeze require
 - ✓ Suitable for light vehicles, motorcycle, scooter etc.
- **Disadvantages**
 - ✓ Low cooling efficiency, noisy due to fan
 - ✓ Not ideal for high-performance, larger engine
 - ✓ Less precise temp control, overheating etc.
- **Used in small (15-20 kW) & Aero-engines**





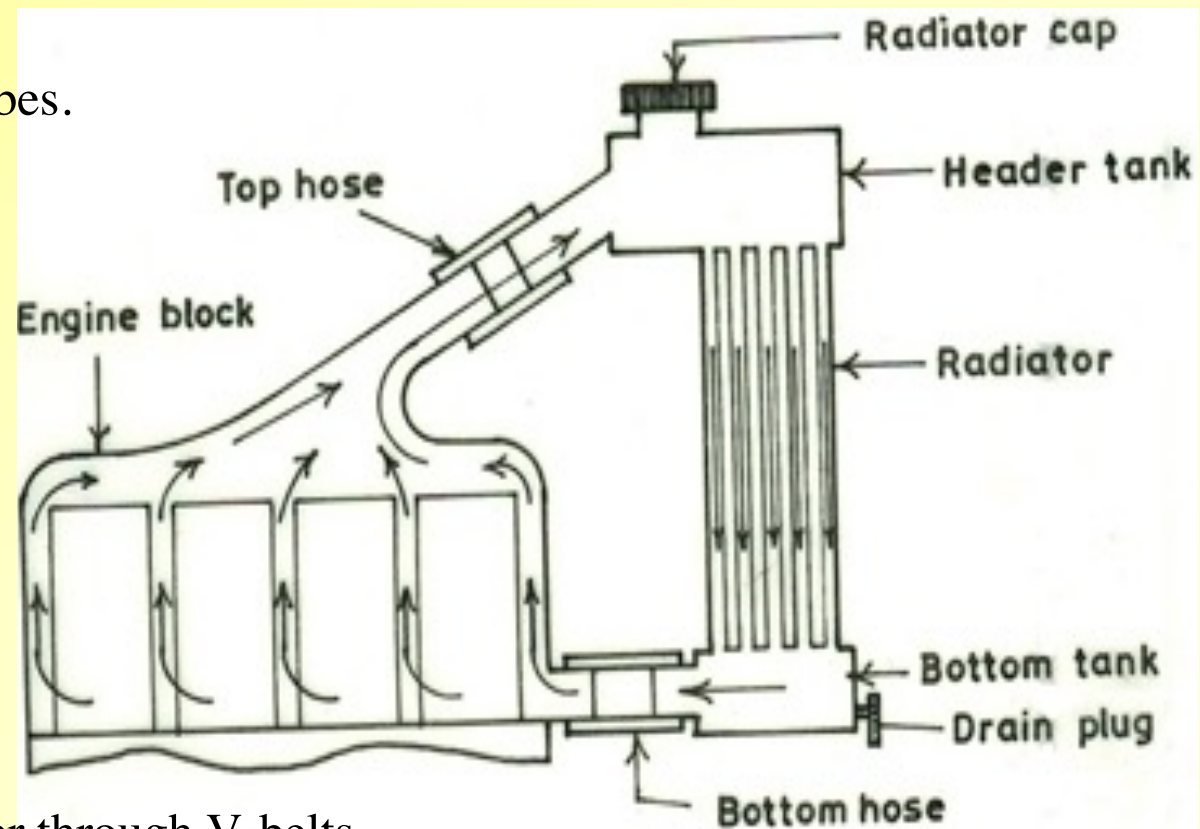
LIQUID COOLING SYSTEMS

➤ *Thermo-syphon Cooling System :*

- ✓ Thermo-syphon cooling is designed in such a way that water circulates naturally because of the **density** difference between the hot water and cold water.
- ✓ So, water pump is not required and water is circulated because of density difference only.
- ✓ Hot water in engine water jacket rises and flows into the upper tank of radiator due to lower density and cold water from radiator enters in water-jacket from lower side to replace hot water.
- ✓ In radiator, hot water from upper tank flow towards lower tank via cooling tubes.
- ✓ To increase heat transfer rate, radiator tubes with fins exposed to air.
- ✓ Thermo-syphon cooling system in automobiles & transformers uses the same principle of cooling.
- ✓ This cooling system is suitable for only low capacity engines

➤ *Pump Circulation System :*

- ✓ In this system circulation of water / liquid is obtained by a pump.
- ✓ Pump is driven by means of shaft power through V-belts.





PUMP CIRCULATION COOLING SYSTEM

- *In this system, water/liquid jackets are provided around the engine cylinder, head, valve seats etc.*
- *Water/fluid in jackets, absorb the heat of combustion.*
- *Hot water/liquid is cooled in radiator by a fan & air flow developed by forward motion of vehicle.*
- *The cooled water/fluid is again recirculated by pump through the water/liquid jackets*
- *Components of water/liquid cooling system*

✓ Radiator: Consists of,

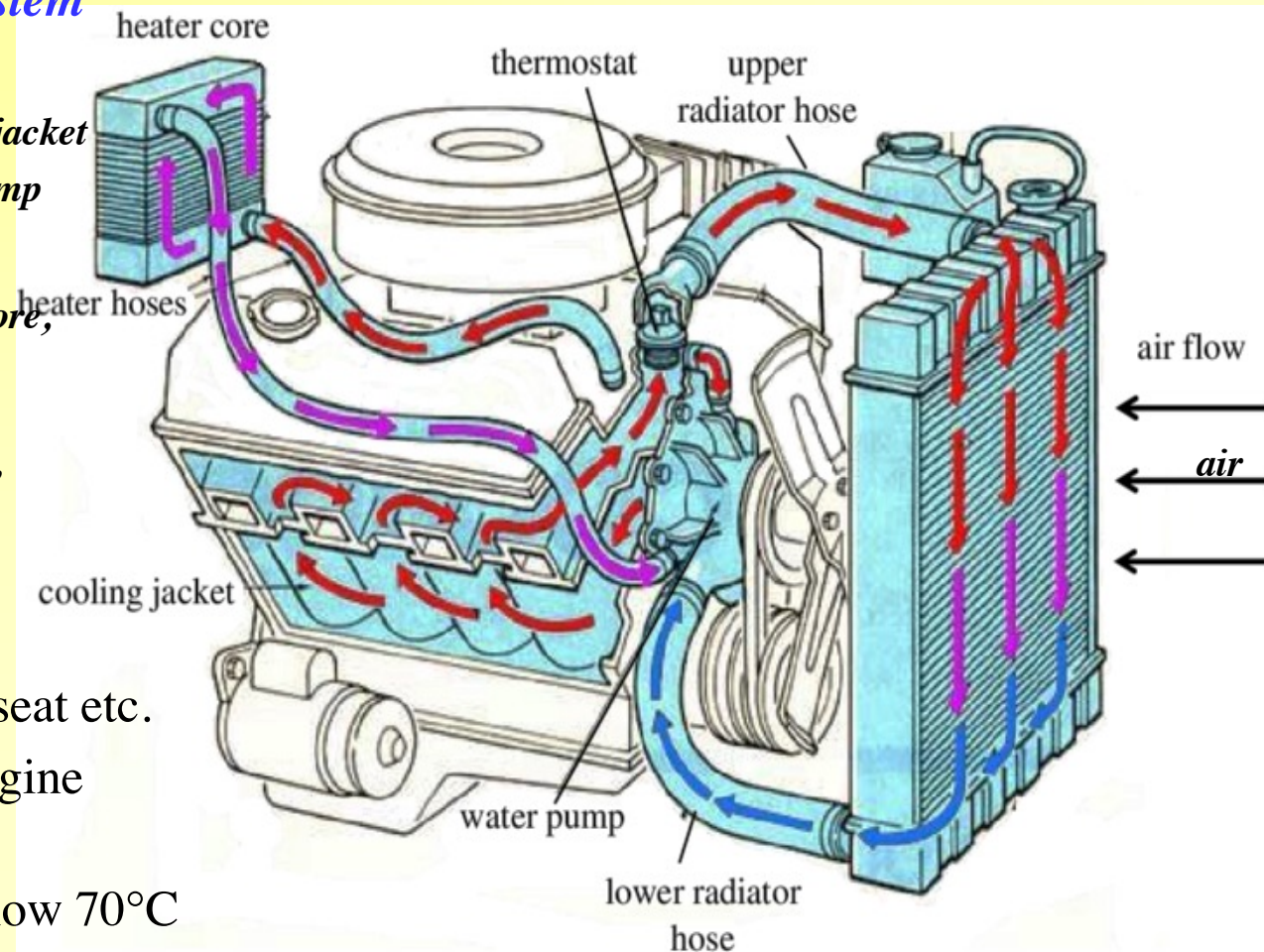
- *Upper tank: collects hot water fluid in jacket*
- *Lower tank: feed liquid in jacket by pump*

✓ Types of radiator cores: **2-types**

- *Tubular: Liquid flow down through core, cooled by fan-air due to forward motion of the vehicle.*
- *Cellular ; Fluid & air flow in passages, and water flowing cool the core*

➤ **Thermostat :**

- ✓ A valve to allow/stop liquid flow from radiator to fluid-jacket, valve seat etc.
- ✓ Auto-function, to attain **efficient** engine operating temperature for best η .
- ✓ Generally, it prevents fluid flow below 70°C

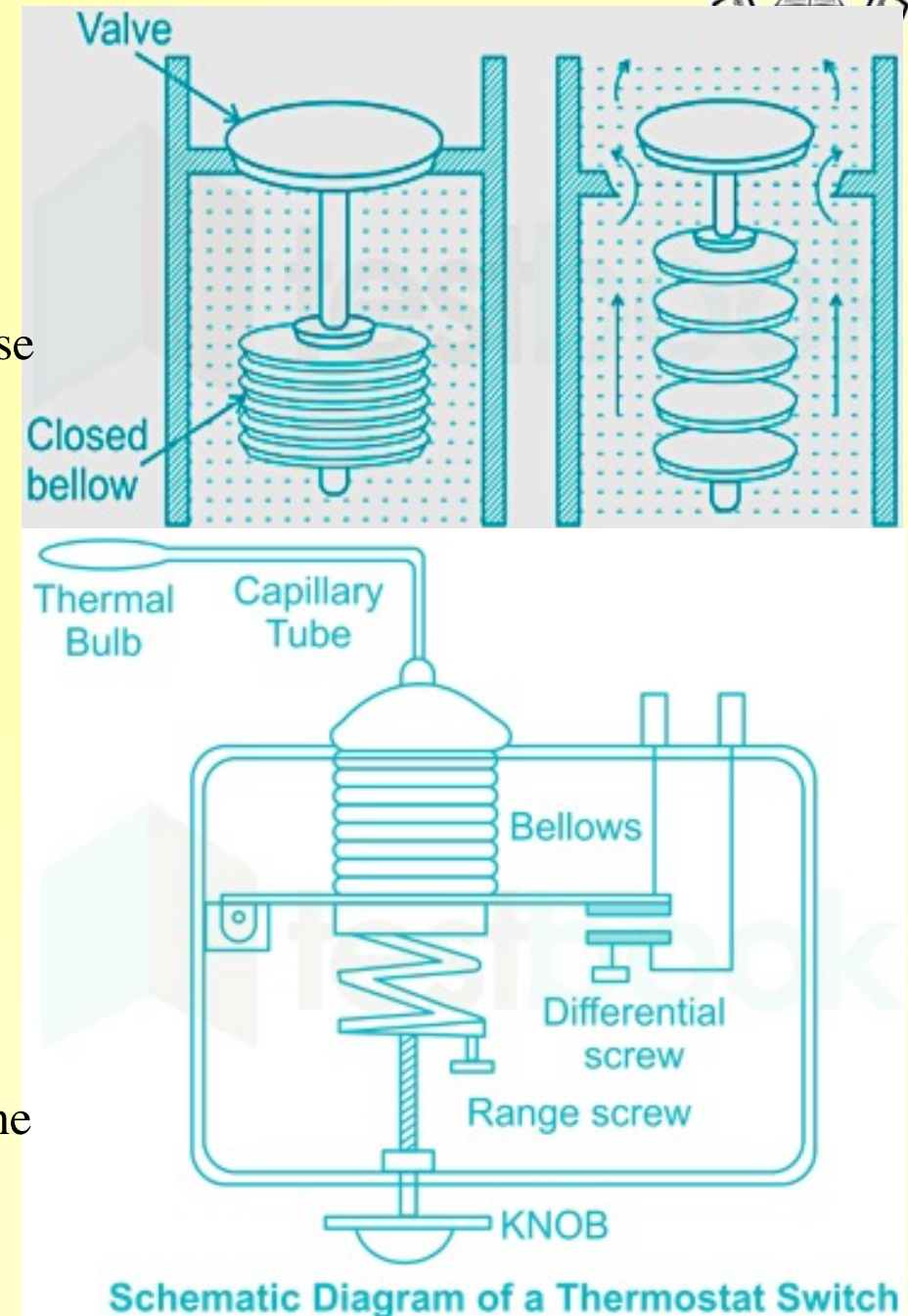




THERMOSTATIC CONTROL



- **Thermostat:**
- **A bellow type thermostatic valves consists of,**
 - ✓ A bronze bellow containing liquid alcohol inside
 - ✓ Bellow operate valve-disc through link for open/close
- **When temperature of water increases,**
 - ✓ The liquid alcohol evaporates and bellow expands
 - ✓ Opens butterfly valve, allows hot water in radiator
- **Advantages**
 - ✓ Uniform cooling of cylinder walls, head & valves
 - ✓ Improves specific fuel consumption, η_{th} of engine
 - ✓ Less noisy as water dampens the engine noise
- **Disadvantages**
 - ✓ Needs liquid / water frequently
 - ✓ Pump consumes auxiliary power of engine,
 - ✓ If cooling system fails cause severe damage to engine
 - ✓ In comparison to others systems, liquid cooled has
 - *More parts, requires frequent maintenance which*
 - *Course a heavy weight engine and system costlier.*





LIQUID COOLING SYSTEM

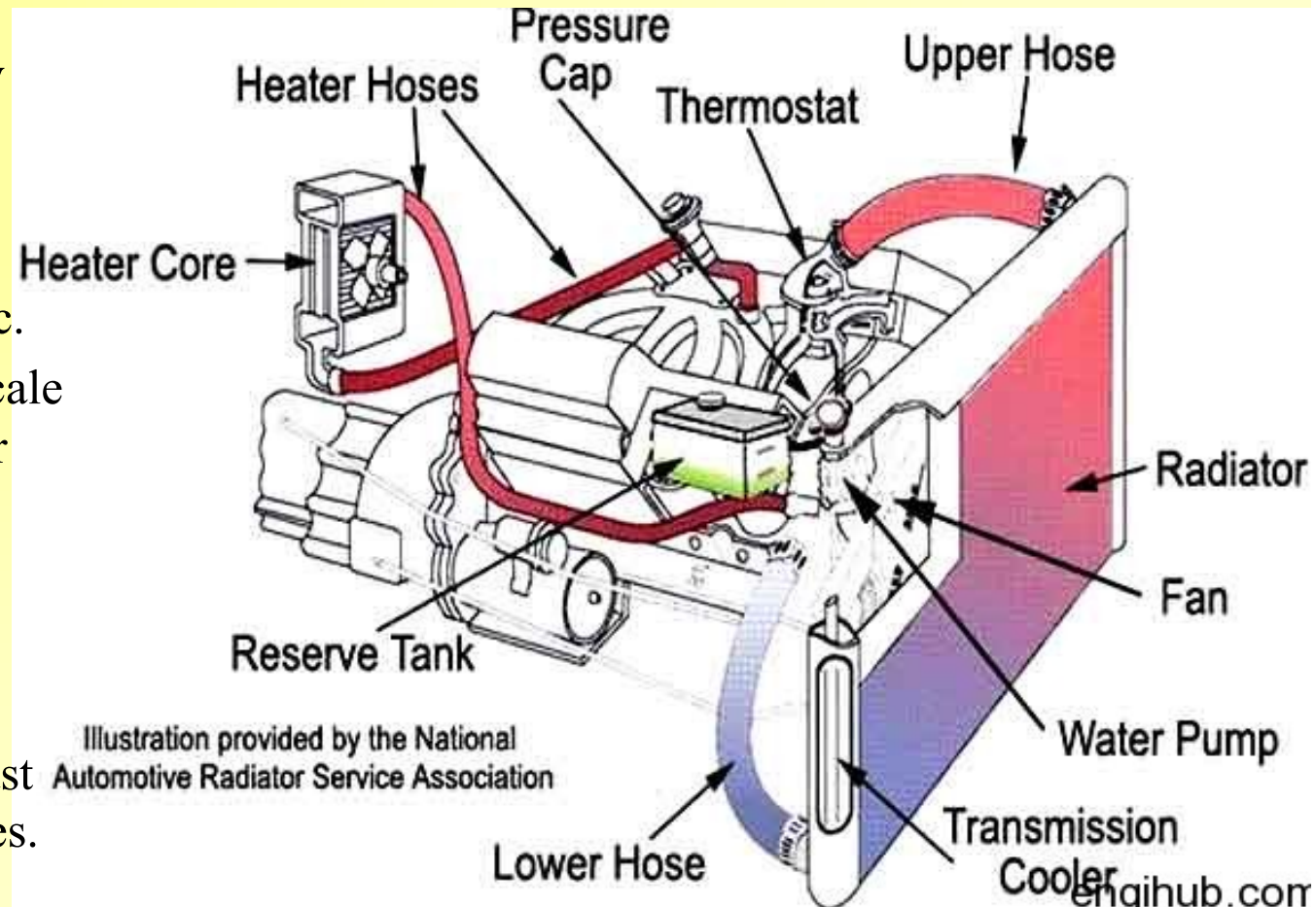
➤ *In water cooling systems, the problems engine faces are*

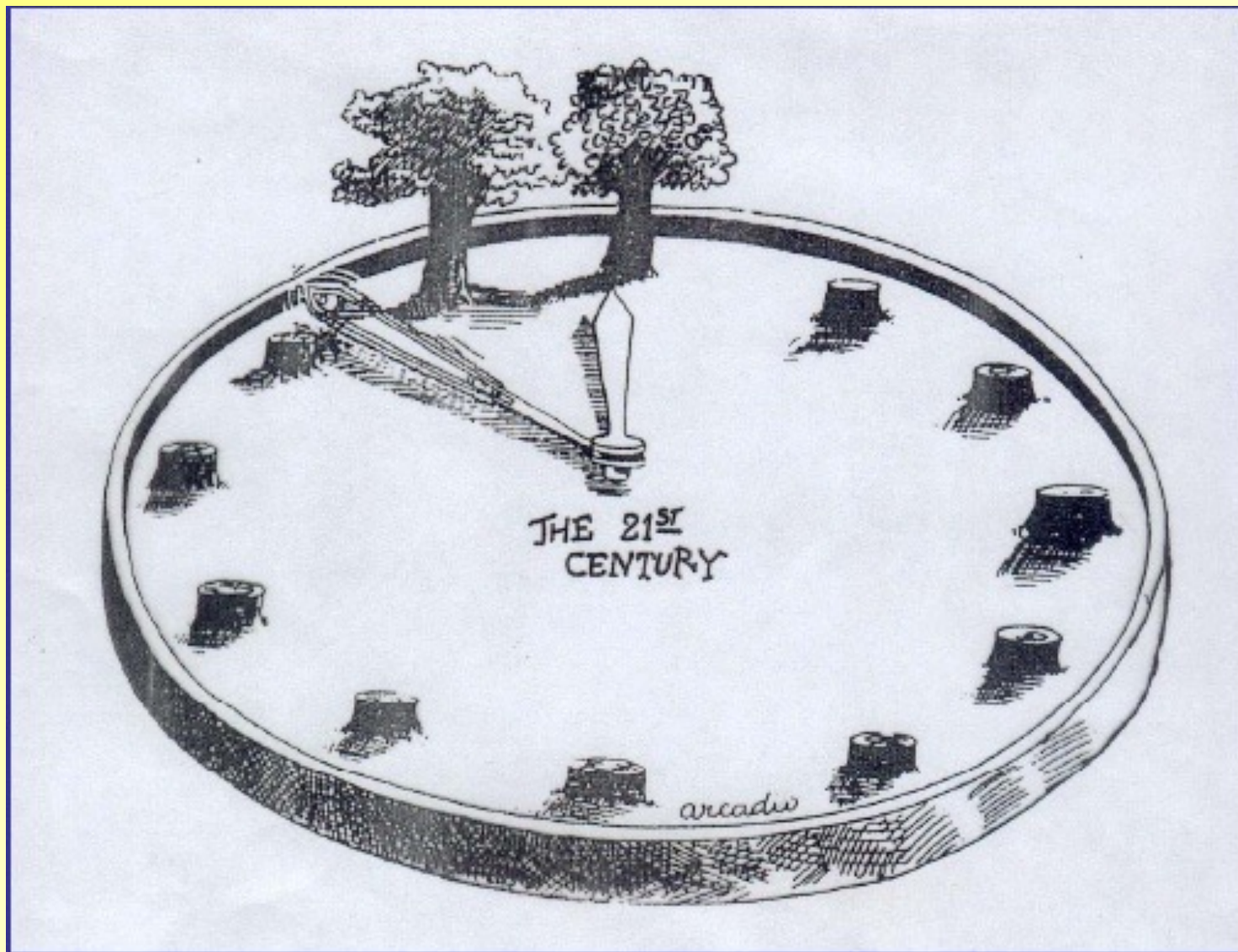
- ✓ Coolant freezing problem at low temp/in cold climates
- ✓ Water causes surface corrosion, gland leakages, boiling at overheating, evaporation loss etc.
- ✓ Evaporation leave rusty trace, scale deposition, reduced heat transfer

➤ *The most automobiles use liquid cooling system.*

➤ *Most liquid-cooled engines use a mixture of water &*

- ✓ Chemicals such as antifreeze, rust inhibitors and other +ve additives.
- ✓ The antifreeze, also known as "engine coolant," are typically glycol-based fluids contain a mixture of ethylene - glycol, additives and water.





Thank You